

5. COMPARATIVE ANALYSIS OF DIFFERENT TYPES OF PROMPTING PATTERNS AND EXPLAIN WITH VARIOUS TEST SCENARIOS

Scenario	Naïve Prompt	Basic Prompt	Observation
Creative Story	Write a story.	Write a 200-word creative story about a student who discovers a hidden library that contains magical books. Include a clear beginning, conflict, and ending.	The naïve prompt generated a generic story, while the basic prompt produced a more detailed, structured, and engaging story.
Factual Question	What is Artificial Intelligence ?	Explain Artificial Intelligence in 150 words. Define AI, describe its key applications, advantages, and provide one real-world example.	The naïve prompt gave a brief definition, whereas the basic prompt provided a comprehensive explanation with examples.
Summarization	Summarize machine learning.	Summarize the concept of machine learning in 100 words, including its definition, types, and major applications.	The basic prompt resulted in a focused and informative summary compared to the shorter response from the naïve prompt.
Advice / Recommendation	Suggest a smartphone.	Recommend a smartphone for a college student with a budget of ₹20,000. Compare battery life, camera quality, performance, and value for money.	The naïve prompt produced a general recommendation, while the basic prompt generated a personalized and detailed comparison.

Analysis Based on Evaluation Criteria

- 1. **Quality**
Basic prompts produced responses that were more organized, relevant, and aligned with the user's requirements. Naïve prompts often resulted in generic outputs with less detail.
- 2. **Accuracy**
Basic prompts improved response accuracy because they clearly specified the expected information and context. Naïve prompts sometimes lacked sufficient context, leading to broader or incomplete answers.
- 3. **Depth**
Responses generated from basic prompts contained more detailed explanations, examples, and structured information. Naïve prompts generally produced shorter and less informative responses.

Comparison Summary

Parameter	Naïve Prompt	Basic Prompt
Clarity	Low	High
Quality of Response	Moderate	High
Accuracy	Moderate	High
Depth of Information	Limited	Detailed
User Satisfaction	Moderate	High